## 710001 POWER UNIT - in panel B2 (factory wired and tested; earthed aluminium enclosure)

**INLET (outside B2)**
powerCON TRUE1 TOP
UL 498, 20 A, locking
v1 position, X 120 Z 313

L N PE

W01
3 x 14
AWG

**BREAKER 1P 15 A**
UL 489, C curve;
protects the 14 AWG
mains wiring

**GFCI RECEPTACLE**
TR duplex NEMA 5-15R,
UL 943 Class A; unit
face, Z 1050 (D-70)

W02 L N

**PE TERMINAL**
inlet PE, receptacle
ground, PS1 FG, earth
stud, B2 frame bond W03

FG

W03 14 AWG green to B2's frame
(only B2 carries a mains part)

**PS1 96 W CLG-100-24**
kit feed, 24 V 4 A, Class 2
input 90-295 V, PFC
FG to the earth stud

F1 5 A
+24 V
0 V PS1

W02 L N

**LAN**
etherCON NE8FDP in B2,
W04 Cat6 to keystone 1,
keystone 2: W31 to the kit

**PS2 60 W LPV-60-24**
24 V 2.5 A, Class 2; light
controller, DMX512 in
Class II (no PE)

PS2 out
F2 5 A
+24 V, UP- DOWN- CALL-

RS-485

W02 L N

**PS3 60 W LPV-60-12**
12 V 5 A, Class 2
input 100-240 V
Class II (no PE)

+12 V, 0 V PS3

port 2

## JOINTS AND HARNESS SEGMENTS (plug pair at every joint; cables clip to the cassettes only, D-26)

X1 unit / B2-CASS
X2 corner, in R1-CASS

W30b 4x16 AWG, 2 per pole B2-CASS
W30c R1-CASS

**X7 VOICE KIT DROP (R1, behind the screen)**
one 24 V Class 2 feed, 5 A fuse: Mini-Fit Jr 4-way, capped
Cat6 keystone, capped; draw wire W32
reached through felt plug FA-KIT

W10b B2-CASS
X4
W10c B4-CASS
X5
W11 in the shelf (6 x 18 AWG feed)

**LIGHT SHELF (back and side wall tops, D-70)**
uplight 4.40 m and down-light 2 x 0.95 m (24 V strips),
call drop at Y -209.5 (X6) down R2-CASS to the bar;
hub keeps up + down + call at about 22 W

W20b air feed 4x18 AWG (2 per pole)
W20c R1-CASS
W20d

W20 and W24 share one 10-way plug at X1, X2 and X3

W24b hub link 6x24 AWG: RS-485, light current, rails
W24c
W24d

W31b Cat6 (RJ45 couplers)
W31c to the kit drop X7

X3 into the air column

### AIR COLUMN (front-right inside corner; hub zone 180 x 180 at Z 1200, D-61, D-70)

**FUSE BLOCK**
behind the hub zone cover
F3a 2 A supply fans
F3b 2 A exhaust fans
F3c 1 A hub

**DESK PRO AI SENSOR HUB**
in-house, cited by name; CO2, presence
out: PWM 1, 2; RS-485 to the lights
in: tach 1, 2; inlet sensor; buttons;
light current; PS1-PS3 rails (H-ELE-05)

**SUPPLY FANS S1 S2**
high in DR; W40 4 x 20 AWG

**EXHAUST FANS E1 E2**
soffit right end; W41

**INLET TEMP SENSOR**
supply fan box inlet; W42

**CONTROLS (hub zone)**
BOOST, TEST, CALL buttons
LIGHT knob: all hub inputs
(W44); TEST = self-test D-87

**CALL LIGHT (R2, above the screen)**
900 mm bar, 2.7 W, on light controller CH3;
fed down from the shelf (X6, W12)
see the light shelf box

**SAFETY AND RESERVES**
SELF-TEST (D-87): TEST checks the 3 rails, runs both fan pairs (tach), steps each light channel (current), reads the sensors and shows PASS or the failing item. PWM fail-safe: an open PWM line runs the fans at full speed. Draw wires: W32 kit drop -> column -> soffit (mic, status light); W33 back wall top -> ceiling cassette (D-41, and the sprinkler or alarm wiring point, ISS-052). Nothing is cut on site later (REQ-VCK-03).

**NOTES**
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Mains (red) exists only inside B2 and the power unit (D-10, IF-12). Nothing mains is connected on site.
2. Every 24 V and 12 V circuit (blue) is Class 2: each supply is a Class 2 power unit on its own sheet (UL 1310), and no two meet (D-168).
3. Plug pairs: supply side receptacle (female), load side plug (male); Molex Mini-Fit Jr, keyed by way count.
4. Every cable through sheet steel runs in a grommet; cables clip to the cassette tray only (D-26).
5. Wire sizes and lengths: harness list (sheet 2 and the _SPEC.csv). Voltage drop under 3 %: calc-results.md.
6. Bought parts are on HOLD: no order, no supplier contact. The in-house hub is cited by name only.
7. Mains: US 120 V 60 Hz, 15 A, UL-listed parts (D-133); the prototype is tested in Vietnam through a 220-to-110 V step-down transformer.
8. Joints: X1 unit/B2-CASS X2 corner (R1-CASS) X3 column entry X4 B2/B4 X5 shelf X6 call X7 kit.

| | |
|---|---|
| PROJECT | FocusPod v2 - Spaceship |
| TITLE | **ELECTRICAL - WIRING DIAGRAM** |
| DWG / PART NO. | 710000 ELECTRICAL |
| REV / STATUS | P01 / DRAFT - not for production |
| UNITS / SCALE | mm / not to scale (schematic) |
| SHEET | 1 of 3 (A1) |
| DATE / BY | 2026-11-01 / Autonomous |
| SOURCE | electrical_model.py, electrical_parts.py |

## HARNESS LIST  (every field, with conductors, routes, bushings and voltage drop: 710000_P01_ELECTRICAL-HARNESS-LIST_SPEC.csv)

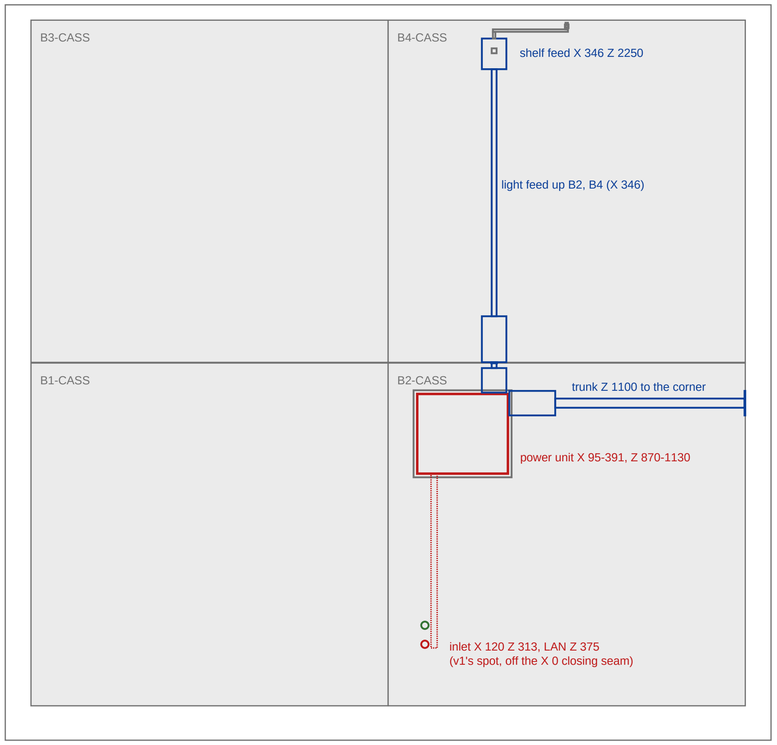
| ID | SEGMENT | FUNCTION | CABLE | L mm | END A | END B |
|---|---|---|---|---|---|---|
| W01 | PU | mains feed | UL 1015 14 AWG single cores (black, white, green) in the sleeve | 920 | inlet terminals (powerCON TRUE1 TOP) | breaker, PE terminal |
| W02 | PU | mains inside the unit | UL 1015 14 AWG (black, white; green to the receptacle) | 1500 | breaker | GFCI receptacle LINE terminals; supply input leads in UL 486C lever connectors |
| W03 | PU | protective bond | UL 1015 14 AWG green | 300 | M6 ring lug, 16-14 AWG | M6 ring lug, 16-14 AWG |
| W04 | PU | LAN in | Cat6 U/UTP patch cord | 870 | RJ45 plug into the feed-through | punched down at keystone jack 1 |
| W10a | PU | light feed | 6 x 18 AWG, Class 2 (CL2 or equivalent) | 300 | light controller terminals | Mini-Fit Jr 6-way receptacle |
| W10b | B2-CASS | light feed | 6 x 18 AWG, Class 2 | 270 | Mini-Fit Jr 6-way plug (mates W10a) | Mini-Fit Jr 6-way receptacle (tail out of the top grommet) |
| W10c | B4-CASS | light feed | 6 x 18 AWG, Class 2 | 1360 | Mini-Fit Jr 6-way plug (mates W10b in P4) | Mini-Fit Jr 6-way receptacle, tail into the shelf |
| W11a | SHELF | light shelf feed lead | 6 x 18 AWG, Class 2 | 300 | Mini-Fit Jr 6-way plug (mates W10c) | strip feeds: back uplight both ways; jumpers W11b, W11c |
| W11b | SHELF | jumper to the left wall uplight | 2 x 18 AWG, Class 2 | 1610 | strip feed connector | strip feed connector |
| W11c | SHELF | jumper to the right wall: uplight, down-light, call drop | 4 x 18 AWG, Class 2 | 1220 | strip feed connector | strip feed connectors; Mini-Fit Jr 2-way receptacle for the call drop (X6) |
| W20a | PU | air feed | 4 x 18 AWG (2 per pole), Class 2 | 300 | PS3 output | Mini-Fit Jr 10-way receptacle (with W24) |
| W20b | B2-CASS | air feed | 4 x 18 AWG (2 per pole), Class 2 | 920 | Mini-Fit Jr 10-way plug (with W24) | Mini-Fit Jr 10-way receptacle (with W24), tail through the corner slot S-CNR |
| W20c | R1-CASS | air feed | 4 x 18 AWG (2 per pole), Class 2 | 1260 | Mini-Fit Jr 10-way plug (with W24) (mated in P2 through the felt slot FS-CNR) | Mini-Fit Jr 10-way receptacle (with W24), tail into the column |
| W30a | PU | voice kit feed | 4 x 16 AWG (2 per pole), Class 2 | 300 | F1, 5 A (PS1, the kit's 24 V feed) | Mini-Fit Jr 4-way receptacle |
| W30b | B2-CASS | voice kit feed | 4 x 16 AWG (2 per pole), Class 2 | 920 | Mini-Fit Jr 4-way plug | Mini-Fit Jr 4-way receptacle, tail through the corner slot S-CNR |
| W30c | R1-CASS | voice kit feed | 4 x 16 AWG (2 per pole), Class 2 | 1050 | Mini-Fit Jr 4-way plug (mated in P2 through the felt slot FS-CNR) | Mini-Fit Jr 4-way receptacle, capped with an empty plug housing (the kit outlet: one 24 V Class 2 feed, two pins per pole, 5 A fuse, 60 W peak) |
| W24a | PU | hub link to the power unit | 6 x 24 AWG screened, Class 2 | 300 | light controller link, PS2 current sense, PS1 and PS2 rail taps | shares W20's 10-way receptacle |
| W24b | B2-CASS | hub link to the power unit | 6 x 24 AWG screened, Class 2 | 920 | shares W20's 10-way plug | shares W20's 10-way receptacle, tail through the corner slot S-CNR |
| W24c | R1-CASS | hub link to the power unit | 6 x 24 AWG screened, Class 2 | 1260 | shares W20's 10-way plug (mated in P2 through the felt slot FS-CNR) | shares W20's 10-way receptacle |
| W31a | PU | voice kit data | Cat6 U/UTP | 300 | punched down at keystone jack 2 (unit face) | RJ45 inline coupler |
| W31b | B2-CASS | voice kit data | Cat6 U/UTP | 920 | RJ45 inline coupler | RJ45 inline coupler, tail through the corner slot S-CNR |
| W31c | R1-CASS | voice kit data | Cat6 U/UTP | 1050 | RJ45 inline coupler (in P2) | RJ45 keystone jack, capped (the kit's data link) |
| W20d | COLUMN | air feed | 4 x 18 AWG (2 per pole), Class 2 | 420 | Mini-Fit Jr 10-way plug (with W24d), mates W20c behind the column | fuse block input |
| W24d | COLUMN | hub link to the power unit | 6 x 24 AWG screened | 420 | shares W20d's 10-way plug | hub I/O (HOLD H-ELE-05) |
| W40 | COLUMN | supply fan pair | 4 x 20 AWG | 1550 | fuse block and hub terminals | 4-way PWM fan connector into a Y-cable at the fan pair (housing HOLD H-ELE-07) |
| W41 | COLUMN | exhaust fan pair | 4 x 20 AWG | 1260 | fuse block and hub terminals | 4-way PWM fan connector into a Y-cable (housing HOLD H-ELE-07) |
| W42 | COLUMN | inlet temperature sensor | 4 x 24 AWG screened | 1750 | hub sensor connector (HOLD, ISS-060) | sensor probe (HOLD H-ELE-06) |
| W43 | COLUMN | hub supply | 2 x 18 AWG, Class 2 | 300 | fuse block | hub power connector (HOLD, ISS-060) |
| W44 | COLUMN | controls loom | 6 x 24 AWG screened | 300 | button and knob tags | hub inputs (HOLD H-ELE-05) |
| W12 | R2-CASS | call light drop | 2 x 18 AWG, Class 2 | 1250 | Mini-Fit Jr 2-way plug (mates W11c's call branch in P7) | strip feed of the call bar |
| W32 | R1-CASS + COLUMN | draw wire: kit zone to the soffit (mic, status light) | nylon draw cord 3 mm | 1870 | tied at P3-KIT | tied at the soffit entry |
| W33 | B4-CASS + CEILING | draw wire: ceiling route (D-41) | nylon draw cord 3 mm | 1680 | tied at P6 | tied at the ceiling cassette's front hole |

**NOTES**
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Lengths: route model (calculation) + 150 mm loops and tails (assumption). The kit drop has a 500 mm service loop.
2. Plug pairs: supply side receptacle (female), load side plug (male). W20 and W24 share one 10-way at X1, X2, X3.
3. Every segment is tested in the factory before it ships (D-87): continuity, polarity, no shorts, latch pull.
4. Cables: Class 2 circuits on 300 V PVC cable (CL2 or CL3 listed for the US phase); mains only inside B2 and the unit.
5. W32 and W33 are draw cords, tied off at each hole, for the voice kit's later cables (REQ-VCK-03).

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|---|---|
| PROJECT | FocusPod v2 - Spaceship |
| TITLE | **ELECTRICAL - HARNESS LIST** |
| DWG / PART NO. | 710000  ELECTRICAL |
| REV / STATUS | P01 / DRAFT - not for production |
| UNITS / SCALE | mm / table, not to scale |
| SHEET | 2 of 3  (A1) |
| DATE / BY | 2026-11-01 / Autonomous |
| SOURCE | electrical_model.py, electrical_parts.py |

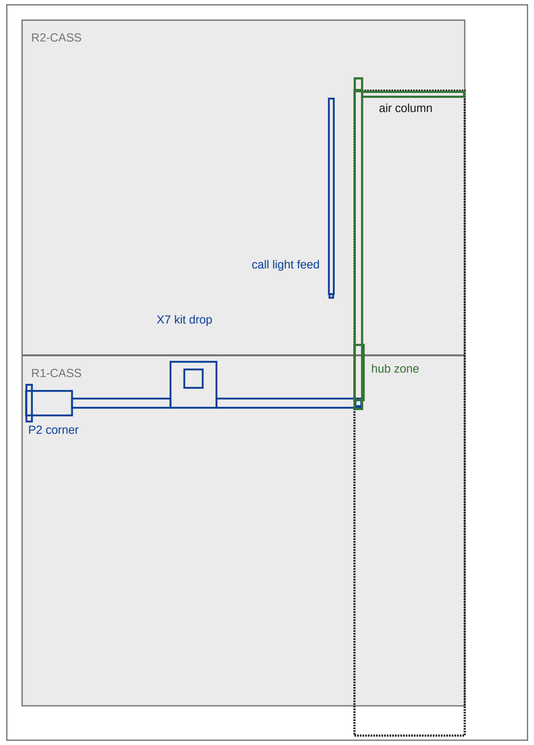
## VIEW A - BACK WALL FROM INSIDE, looking toward +Y (1:10)

X view = +X, Y view = +Z; origin O on the room floor at the pod centre




## VIEW B - RIGHT WALL FROM INSIDE, looking toward +X (1:10)

X view = -Y (front to the right), Y view = +Z




**RESERVED SPACES IN THE CAD**

Routes, pockets, bushings and slots are reserved spaces in the cassettes' 24 mm gap, in 710000_P01_ELECTRICAL_CAD, derived from skeleton P04. Pockets: P1 at the unit, P2 in R1-CASS behind the felt slot FS-CNR (hidden by B2-CASS's end), P4 above the B2/B4 joint, P5 and P6 the light feed, P3 the voice kit drop.

**SITE ORDER (two people, one 5 mm key, no electrician)**

1 Hang R1-CASS and R2-CASS; their tails stay out.
2 Hold B2-CASS near: plug its leads to the unit (X1) and to R1's corner tails (X2), push the pairs into P1 and P2, hang it.
3 B4-CASS: plug X4, push the pair into P4, hang it.
4 Plug the light shelf at X5 when the shelf goes up.
5 Before the air column goes in, plug X3 and X6 behind it.
6 Plug the fans at their boxes (DR inner face; soffit underside).
7 Room cord into the inlet, breaker on, press TEST: PASS (D-87).

**KEEP-OUTS FOR OTHER TEAMS**

WALL: B2 bay X 95-391, Z 870-1130 for the unit; sleeve X 140-160, Z 301-870; the holes and slots in B2-CASS, B4-CASS, R1-CASS, R2-CASS.
INTERIOR: ledge and screen fixings 15 mm off the trunk band Z 1085-1115.
CEILING: column chase 24 x 24 (X 1140-1164, Y -310 to -286), hub zone, soffit entry; the ceiling cassette's holes at X 583 (D-41 route).
FRONT: a 19.1 mm bushed hole in DR's back plate, X 1152, Z 2108 (G-DR).

ISO first angle

**NOTES**
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Every coordinate in these views is computed from electrical_model.py (skeleton P04 parameters), so they follow the CAD.
2. Bushings (Heyco 2836, 19.1 hole): the unit's H01-H03, G-B2-B4 x2, G-CEIL-B/F, G-DR (DR back plate), G-SOFFIT; the slot S-CNR gets an edge strip.
3. Felt-only holes (FX-*, FS-CNR, FA-KIT) go through the felt only, clear of the 12 mm lips: no steel edge, no bushing (no hole through a lip, D-69).
4. These views are the route drawing of 710000; there is no separate Fusion drawing of the route model (PM ruling 2026-11-01).

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|---|---|
| PROJECT | FocusPod v2 - Spaceship |
| TITLE | **ELECTRICAL - ROUTES AND JOINTS** |
| DWG / PART NO. | 710000 ELECTRICAL |
| REV / STATUS | P01 / DRAFT - not for production |
| UNITS / SCALE | mm / 1:10 (views) |
| SHEET | 3 of 3 (A1) |
| DATE / BY | 2026-11-01 / Autonomous |
| SOURCE | electrical_model.py, electrical_parts.py |